

Growth rates of modern science:

A bibliometric analysis based on the number of publications and cited references

Lutz Bornmann¹, Rüdiger Mutz²

¹ Corresponding author: Division for Science and Innovation Studies, Administrative
Headquarters of the Max Planck Society, Munich, Germany, Email: bornmann@gv.mpg.de,
Tel: +49 89 2108 1265

² Professorship for Social Psychology and Research on Higher Education, ETH Zurich,
Zurich, Switzerland, Email: ruediger.mutz@gess.ethz.ch, Tel: +41 44 632 4918

Abstract

Many studies in information science have looked at the growth of science. In this study, we re-examine the question of the growth of science. To do this we (i) use current data up to publication year 2012 and (ii) analyse it across all disciplines and also separately for the natural sciences and for the medical and health sciences. Furthermore, the data are analysed with an advanced statistical technique – segmented regression analysis – which can identify specific segments with similar growth rates in the history of science. The study is based on two different sets of bibliometric data: (1) The number of publications held as source items in the Web of Science (WoS, Thomson Reuters) per publication year and (2) the number of cited references in the publications of the source items per cited reference year. We have looked at the rate at which science has grown since the mid-1600s. In our analysis of cited references we identified three growth phases in the development of science, which each led to growth rates tripling in comparison with the previous phase: from less than 1% up to the middle of the 18th century, to 2 to 3% up to the period between the two world wars and 8 to 9% to 2012.

Key words

growth of science; bibliometrics; cited references

1 Introduction

Many studies in information science have looked at the growth of science (Evans, 2013). Tabah (1999) offers an overview of the literature which groups these studies under the label "the study of literature dynamics" (p. 249): "The information science approach is to follow the published literature and infer from the growth of the literature the movement of ideas and associations between scientists" (Tabah, 1999, p. 249). Price (1965; 1951, 1961) can undoubtedly be seen as a pioneering researcher on literature dynamics (de Bellis, 2009). Price analysed the references listed in the 1961 edition of the Science Citation Index (SCI, Thomson Reuters) and the papers collected in the *Philosophical Transactions of the Royal Society of London*. His results show that science is growing exponentially (in a certain period by a certain percentage rate) and doubles in size every 10 to 15 years. The exponential growth in science established by Price has become today a generally accepted thesis which has also been confirmed by other studies (Tabah, 1999).

In this study, we want to re-examine the question of the growth of science. To do this we will (i) use current data up to publication year 2012 and (ii) analyse it across all disciplines and also separately for the natural sciences and for the medical and health sciences. Furthermore, the data will be analysed with an advanced statistical technique – segmented regression analysis – which can identify specific segments with similar growth rates in the history of science. The study is based on two different sets of data: (1) The number of publications held as source items in the Web of Science (WoS, Thomson Reuters) per publication year and (2) the number of cited references in the publications of the source items per cited reference year (Bornmann & Marx, 2013; Marx, Bornmann, Barth, & Leydesdorff, in press). The advantage of using cited references rather than source items is that they can give insight into the early period of modern science. There is no database available which covers publications (source items) from the early period. The disadvantage of using cited

references is that the literature which has not been cited yet is not considered. Furthermore, publishing in the early period is inferred by today's citing (here: in the period from 1980 to 2012).

2 Methods

Publications are a very suitable source of data with which to investigate the growth rates of science: "Communication in science is realized through publications. Thus, scientific explanations, and in general scientific knowledge, are contained in written documents constituting scientific literature" (Riviera, 2013, p. 1446). Having a paper published in a journal is an integral part of being a scientist: "[It] is a permanent record of what has been discovered, when and by which scientists – like a court register for science – [and it] shows the quality of the scientist's work: other experts have rated it as valid, significant and original" (Sense About Science, 2005). Because "efficient research requires awareness of all prior research and technology that could impact the research topic of interest, and builds upon these past advances to create discovery and new advances" (Kostoff & Shlesinger, 2005, p. 199), cited references in the publications are also an important source of data with which to examine scientific growth. An increase in the number of cited references indicates that there are more citing and/or cited publications.

Our study is based on all the publications from 1980 to 2012 and the cited references in these publications. The data is taken from an in-house database belonging to the Max Planck Society (Munich, Germany) based on WoS. It was established and is maintained by the Max Planck Digital Library (MPDL, Munich, Germany). As the data prepared by the MPDL relates to publications (and their cited references) since 1980, it was only possible to include these publications (and their cited references) in the analysis. The first step in the study was to select all the publications (all document types) that appeared between 1980 and 2012 (38,508,986 publications) and determine the number of publications per year. The